

Problem 3 (Bulgarian solitaire), Cell 4: one above a triangular number

Problem 3 team

Official setting.

A partition of a positive integer n is a weakly decreasing sequence $\lambda = (\lambda_1, \dots, \lambda_s)$ of positive integers with $\lambda_1 + \dots + \lambda_s = n$. The shift $\mathcal{B}(\lambda)$ is the partition whose parts are the positive numbers among $\lambda_1 - 1, \dots, \lambda_s - 1$ together with one extra part equal to s . Call λ cyclic if $\mathcal{B}^i(\lambda) = \lambda$ for some $i \geq 1$, and let $d_{\mathcal{B}}(\lambda) = \min\{i \geq 0 : \mathcal{B}^i(\lambda) \text{ is cyclic}\}$, $D_{\mathcal{B}}(n) = \max\{d_{\mathcal{B}}(\lambda) : \lambda \text{ is a partition of } n\}$. Write $T_k = k(k+1)/2$ and $\delta_k = (k, k-1, \dots, 2, 1)$. Every $n \geq 1$ satisfies $T_{k-1} < n \leq T_k$ for exactly one k , which we call the rank of n .

Section 1 restates these definitions in an equivalent form, proves that $d_{\mathcal{B}}$ is well defined and that the rank exists and is unique (Lemma 1.2, where we also write $n = T_{k-1} + r$, $1 \leq r \leq k$), and fixes further notation.

Official statement of the cell.

$n = T_{k-1} + 1$, that is $n = 11, 16, 22, 29, \dots$ for $k = 5, 6, 7, 8, \dots$. Determine $D_{\mathcal{B}}(T_{k-1} + 1)$ for every $k \geq 5$, with proof of both bounds.

Status. Solved (complete proof; no computation is used). **New relative to Griggs–Ho:** this is the case $r = 1$ of their Conjecture 4.7 [4], which they left open (they confirmed it only for $n \leq 36$ and for $r \in \{k-1, k\}$). *Caveat on novelty:* we checked the sources cited here and three later ones (Drensky’s 2015 survey, the 2025 paper of Harris–Nguyen, and a 2026 survey). None of them states this case, but we did not survey the literature exhaustively, so it may already be known.

Answer and where it is proved.

- $D_{\mathcal{B}}(T_{k-1} + 1) = (k-1)(k-3)$ for every $k \geq 5$ (Corollary 7.3).
- Upper bound: Theorem 6.1, a single argument for all $k \geq 5$. The new ingredient is the *entry lemma* (Lemma 5.2), proved for all $k \geq 3$: the last non-cyclic partition of every orbit has exactly $k-2$ parts, and the orbit never enters the cycle at the 2nd or the k -th cyclic partition.
- Lower bound: $\lambda^{\{k\}} = (k-2, k-2, k-3, k-4, \dots, 3, 2, 2, 1)$ ($\lambda_1 = k-2$, $\lambda_i = k-i$ for $2 \leq i \leq k-2$, $\lambda_{k-1} = 2$, $\lambda_k = 1$) has $d_{\mathcal{B}} = (k-1)(k-3)$ for $k \geq 4$ (Theorem 7.2; this is the witness of [4, Thm. 4.5, case (1)] with $r = 1$).
- In addition: all extremal partitions are characterised in Theorem 8.1, and Section 9 explains where the Cell 3 argument loses.

Sections 1–4 are shared with Cells 2, 3 and 5, and are repeated so that this document is self-contained.

1 Setting and notation

Throughout, $T_m = m(m+1)/2$ for integers $m \geq 0$, and $[P] \in \{0, 1\}$ denotes the truth value of a statement P . A *partition* λ of $n \geq 1$ (written $\lambda \vdash n$) is a finite non-increasing sequence of positive integers $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_\ell$ with sum n ; $\ell = \ell(\lambda)$ is its number of parts, and we put $\lambda_i = 0$ for $i > \ell(\lambda)$. Partitions are identified with multisets of positive integers.

Definition 1.1 (Bulgarian solitaire). For $\lambda \vdash n$ let $\mathcal{B}(\lambda)$ be the partition whose parts are $\ell(\lambda)$ together with the numbers $\lambda_i - 1$ for those $i \leq \ell(\lambda)$ with $\lambda_i \geq 2$. Since $\ell(\lambda) + \sum_i (\lambda_i - 1) = n$, we have $\mathcal{B}(\lambda) \vdash n$. Put $\mathcal{B}^0 = \text{id}$ and $\mathcal{B}^{i+1} = \mathcal{B} \circ \mathcal{B}^i$. A partition μ is *cyclic* if $\mathcal{B}^P(\mu) = \mu$ for some $P \geq 1$. For $\lambda \vdash n$, $d_{\mathcal{B}}(\lambda)$ is the least $i \geq 0$ such that $\mathcal{B}^i(\lambda)$ is cyclic, and $D_{\mathcal{B}}(n) = \max\{d_{\mathcal{B}}(\lambda) : \lambda \vdash n\}$. A partition $\lambda \vdash n$ with $d_{\mathcal{B}}(\lambda) = D_{\mathcal{B}}(n)$ is called *extremal*.

$d_{\mathcal{B}}(\lambda)$ is well defined: n has finitely many partitions, so $\mathcal{B}^a(\lambda) = \mathcal{B}^{a+P}(\lambda)$ for some $a \geq 0$, $P \geq 1$, and then $\mathcal{B}^a(\lambda)$ is cyclic. If μ is cyclic with $\mathcal{B}^P(\mu) = \mu$, then $\mathcal{B}^P(\mathcal{B}(\mu)) = \mathcal{B}(\mu)$, so $\mathcal{B}(\mu)$ is cyclic. Hence $\mathcal{B}^i(\lambda)$ is cyclic for every $i \geq d_{\mathcal{B}}(\lambda)$ and for no $i < d_{\mathcal{B}}(\lambda)$.

Lemma 1.2 (rank). *Every $n \geq 1$ can be written in exactly one way as $n = T_{k-1} + r$ with integers $k \geq 1$ and $1 \leq r \leq k$. We call k the rank of n ; n is triangular iff $r = k$ (then $n = T_k$).*

Proof. $T_0 = 0 < T_1 < T_2 < \dots$ is strictly increasing and unbounded, so there is exactly one $k \geq 1$ with $T_{k-1} < n \leq T_k$; then $r = n - T_{k-1}$ satisfies $1 \leq r \leq T_k - T_{k-1} = k$. Conversely $n = T_{k-1} + r$ with $1 \leq r \leq k$ forces $T_{k-1} < n \leq T_k$, i.e. this k . If $n = T_m$ ($m \geq 1$), then $T_{m-1} < n \leq T_m$, so $k = m$ and $r = T_m - T_{m-1} = m = k$; conversely $r = k$ gives $n = T_k$. $\square$

2 Diagrams, diagonals, rotation, and Brandt's theorem

This section follows Griggs–Ho [4, §2] (the idea goes back to Etienne [3]); we make their “left shifting decreases the diagonal index” precise with an explicit potential Φ .

Let $\mathbb{P} = \mathbb{Z}_{>0} \times \mathbb{Z}_{>0}$; the *cell* $(i, j) \in \mathbb{P}$ has *row* i and *column* j . The *diagram* of λ is

$$Y(\lambda) = \{(i, j) \in \mathbb{P} : 1 \leq j \leq \ell(\lambda), 1 \leq i \leq \lambda_j\},$$

so column j has λ_j cells (columns are parts, as in [4]). $Y(\lambda)$ determines λ , since $\lambda_j = \#\{i : (i, j) \in Y(\lambda)\}$. Row 1 of $Y(\lambda)$ is $\{(1, j) : j \leq \ell(\lambda)\}$ and column 1 has λ_1 cells; in particular

$$(1, w+1) \in Y(\lambda) \implies (1, w) \in Y(\lambda) \quad (w \geq 1). \quad (1)$$

For a finite sequence $a = (a_1, \dots, a_m)$ of non-negative integers let $X_a = \{(i, j) : 1 \leq j \leq m, 1 \leq i \leq a_j\}$. If a is non-increasing, then $X_a = Y(\nu)$ where ν is the partition formed by the non-zero entries of a .

For $w \geq 1$ the w -th *diagonal* is $D_w = \{(i, j) \in \mathbb{P} : i + j = w + 1\}$; it has the w cells $d_w(j) = (w + 1 - j, j)$, $1 \leq j \leq w$ ($d_w(j)$ is the cell of D_w in column j). $\mathbb{P}$ is the disjoint union of the D_w . Put $\Delta_0 = \emptyset$ and $\Delta_m = D_1 \cup \dots \cup D_m$ for $m \geq 1$. Column j of Δ_m consists of the cells (i, j) with $i \leq m + 1 - j$, so $\Delta_m = Y(\delta_m)$ with $\delta_m = (m, m-1, \dots, 1)$, and $|\Delta_m| = T_m$.

Define $\rho : \mathbb{P} \rightarrow \mathbb{P}$ by $\rho(i, j) = (i-1, j+1)$ if $i \geq 2$ and $\rho(1, j) = (j, 1)$.

Lemma 2.1. (a) $\rho(d_w(j)) = d_w(j+1)$ for $1 \leq j < w$ and $\rho(d_w(w)) = d_w(1)$. Hence ρ maps each D_w bijectively onto itself, ρ is a bijection of $\mathbb{P}$, and for $m \geq 0$, $\rho^m(d_w(j)) = d_w(j')$ where $1 \leq j' \leq w$ and $j' \equiv j + m \pmod{w}$. In particular ρ^w is the identity on D_w . (b) ρ preserves the quantity $i + j$ of a cell.

Proof. (a) For $j < w$ the cell $d_w(j) = (w+1-j, j)$ has row $w+1-j \geq 2$, so $\rho(d_w(j)) = (w-j, j+1) = d_w(j+1)$; and $\rho(d_w(w)) = \rho(1, w) = (w, 1) = d_w(1)$. So ρ acts on D_w as the cyclic shift $j \mapsto j+1$ of column indices modulo w ; the rest follows. (b) $(i-1) + (j+1) = i+j$ and $j+1 = 1+j$. $\square$

Lemma 2.2 (one move). *Let $\lambda \vdash n$, $s = \ell(\lambda)$ and $a = (s, \lambda_1 - 1, \lambda_2 - 1, \dots, \lambda_s - 1)$. Then $\rho(Y(\lambda)) = X_a$, and $Y(\mathcal{B}(\lambda)) = X_{a^*}$, where a^* is the non-increasing rearrangement of a . If $s \geq \lambda_1 - 1$, then a is non-increasing and $Y(\mathcal{B}(\lambda)) = \rho(Y(\lambda))$.*

Proof. The cells of row 1 of $Y(\lambda)$ are $(1, j)$, $1 \leq j \leq s$; ρ sends them to $(j, 1)$, i.e. exactly onto the cells $(1, 1), \dots, (s, 1)$ of column 1. A cell $(i, j) \in Y(\lambda)$ with $i \geq 2$ satisfies $2 \leq i \leq \lambda_j$ and is sent to $(i-1, j+1)$; for fixed j these images are exactly the cells $(1, j+1), \dots, (\lambda_j - 1, j+1)$. Hence $\rho(Y(\lambda)) = X_a$. The parts of $\mathcal{B}(\lambda)$ are the non-zero entries of a , so $Y(\mathcal{B}(\lambda)) = X_{a^*}$. The entries of a after the first are non-increasing because λ is; so a is non-increasing iff $s \geq \lambda_1 - 1$, and then $a^* = a$. $\square$

For a finite set $X \subseteq \mathbb{P}$ put $\Phi(X) = \sum_{(i,j) \in X} (i+j)$.

Lemma 2.3. *For every finite sequence a of non-negative integers, $\Phi(X_{a^*}) \leq \Phi(X_a)$, with equality iff a is non-increasing.*

Proof. $\Phi(X_a) = \sum_j \sum_{i=1}^{a_j} (i+j) = \sum_j \binom{a_j+1}{2} + F(a)$ with $F(a) = \sum_j j a_j$. The first sum does not change when a is permuted. If $a_j < a_{j+1}$, exchanging these two entries changes F by $j a_{j+1} + (j+1) a_j - j a_j - (j+1) a_{j+1} = a_j - a_{j+1} < 0$. Starting from a , repeatedly exchange some adjacent pair with $a_j < a_{j+1}$. Each exchange lowers the non-negative integer F , so the process stops, and it stops at a sequence with no adjacent ascent, i.e. at the non-increasing rearrangement a^* (which is unique). If a is not non-increasing at least one exchange happens, so $F(a^*) < F(a)$; if a is non-increasing then $a = a^*$. $\square$

Corollary 2.4. *For $\lambda \vdash n$: $\Phi(Y(\mathcal{B}(\lambda))) \leq \Phi(Y(\lambda))$, with equality iff $\ell(\lambda) \geq \lambda_1 - 1$; in the case of equality $Y(\mathcal{B}(\lambda)) = \rho(Y(\lambda))$.*

Proof. By Lemma 2.1, ρ is injective and preserves $i+j$, so $\Phi(\rho(X)) = \Phi(X)$. With a as in Lemma 2.2: $\Phi(Y(\lambda)) = \Phi(X_a) \geq \Phi(X_{a^*}) = \Phi(Y(\mathcal{B}(\lambda)))$, with equality iff a is non-increasing (Lemma 2.3), i.e. iff $\ell(\lambda) \geq \lambda_1 - 1$, and then Lemma 2.2 gives $Y(\mathcal{B}(\lambda)) = \rho(Y(\lambda))$. $\square$

For $k \geq 1$ and $E \subseteq D_k$ let λ_E be the partition whose j -th part, $1 \leq j \leq k$, is $k-j + [d_k(j) \in E]$ (zero parts omitted). The sequence $(k-j + \varepsilon_j)_{j=1}^k$ with $\varepsilon_j \in \{0, 1\}$ is non-increasing (consecutive differences are $1 + \varepsilon_j - \varepsilon_{j+1} \geq 0$), so λ_E is a partition, $|\lambda_E| = T_{k-1} + |E|$, and $Y(\lambda_E) = \Delta_{k-1} \cup E$ (column j consists of the $k-j$ cells of Δ_{k-1} and, if $d_k(j) \in E$, the cell $d_k(j) = (k+1-j, j)$). Writing $e_{k-j} = [d_k(j) \in E]$ we get $\lambda_E = (k-1 + e_{k-1}, k-2 + e_{k-2}, \dots, 1 + e_1, e_0)$.

Theorem 2.5 (Brandt [2]; see also [1, 3, 4]). *Let $n = T_{k-1} + r$ with $1 \leq r \leq k$. A partition of n is cyclic if and only if it equals λ_E for some $E \subseteq D_k$ with $|E| = r$; equivalently, iff it has the form $(k-1 + e_{k-1}, \dots, 1 + e_1, e_0)$ with $e_i \in \{0, 1\}$ and $\sum_i e_i = r$ (zero parts omitted). Moreover $Y(\mathcal{B}(\lambda_E)) = \rho(Y(\lambda_E))$, i.e. $\mathcal{B}(\lambda_E) = \lambda_{\rho(E)}$, and $\mathcal{B}^k(\lambda_E) = \lambda_E$.*

Proof. ($\Leftarrow$ and “moreover”). Let $\mu = \lambda_E$ with $E \subseteq D_k$, $|E| = r$. Row 1 of $Y(\mu) = \Delta_{k-1} \cup E$ consists of the cells $(1, w) = d_w(w)$ with $w \leq k-1$ and possibly $d_k(k)$; so $\ell(\mu) = k-1 + [d_k(k) \in E] \geq k-1$. Column 1 consists of $d_w(1)$ for $w \leq k-1$ and possibly $d_k(1)$; so $\mu_1 = k-1 + [d_k(1) \in E] \leq k$. Thus $\ell(\mu) \geq \mu_1 - 1$ and Corollary 2.4 and Lemma 2.1 give $Y(\mathcal{B}(\mu)) = \rho(\Delta_{k-1} \cup \rho(E)) = \Delta_{k-1} \cup \rho(E)$ with $\rho(E) \subseteq D_k$, $|\rho(E)| = r$. Hence $\mathcal{B}(\lambda_E) = \lambda_{\rho(E)}$, and by induction $\mathcal{B}^k(\lambda_E) = \lambda_{\rho^k(E)} = \lambda_E$ because ρ^k is the identity on D_k . So μ is cyclic.

($\Rightarrow$). Let $\lambda \vdash n$ with $\mathcal{B}^P(\lambda) = \lambda$, $P \geq 1$, and write $\lambda^{(m)} = \mathcal{B}^m(\lambda)$. By Corollary 2.4, $\Phi(Y(\lambda^{(0)})) \geq \Phi(Y(\lambda^{(1)})) \geq \dots \geq \Phi(Y(\lambda^{(P)})) = \Phi(Y(\lambda^{(0)}))$, so all these are equalities and $Y(\lambda^{(m+1)}) = \rho(Y(\lambda^{(m)}))$ for $0 \leq m < P$. As $\lambda^{(m+P)} = \lambda^{(m)}$, this holds for all $m \geq 0$, and therefore $Y(\lambda^{(m)}) = \rho^m(Y(\lambda))$ for all $m \geq 0$.

Claim: there is no $w \geq 1$ such that $D_w \not\subseteq Y(\lambda)$ and $D_{w+1} \cap Y(\lambda) \neq \emptyset$. Suppose $d_w(\alpha) \notin Y(\lambda)$ and $d_{w+1}(\beta) \in Y(\lambda)$. Since $\gcd(w, w+1) = 1$, the Chinese remainder theorem gives $m \geq 0$ with $m \equiv w - \alpha \pmod{w}$ and $m \equiv w + 1 - \beta \pmod{w+1}$. By Lemma 2.1, $\rho^m(d_w(\alpha)) = d_w(w) = (1, w)$ and $\rho^m(d_{w+1}(\beta)) = d_{w+1}(w+1) = (1, w+1)$. As ρ^m is injective, $(1, w) \notin \rho^m(Y(\lambda)) = Y(\lambda^{(m)})$ while $(1, w+1) \in Y(\lambda^{(m)})$, contradicting (1).

$Y(\lambda)$ is finite, so some D_w is not contained in it; let $K \geq 1$ be the least such w . Then $\Delta_{K-1} \subseteq Y(\lambda)$. By the claim, $D_{K+1} \cap Y(\lambda) = \emptyset$; then $D_{K+1} \not\subseteq Y(\lambda)$, so again by the claim $D_{K+2} \cap Y(\lambda) = \emptyset$, and inductively $D_w \cap Y(\lambda) = \emptyset$ for all $w > K$. Thus $Y(\lambda) = \Delta_{K-1} \cup E'$ with $E' = Y(\lambda) \cap D_K \neq D_K$; let $r' = |E'| \in \{0, \dots, K-1\}$, so $n = T_{K-1} + r'$ and $Y(\lambda) = Y(\lambda_{E'})$ (with k replaced by K in the definition of $\lambda_{E'}$). If $r' \geq 1$, then $T_{K-1} < n < T_K$, so by Lemma 1.2 $K = k$, $r' = r$, and $\lambda = \lambda_{E'}$. If $r' = 0$, then $K \geq 2$ (as $n \geq 1$) and $n = T_{K-1} = T_{K-2} + (K-1)$, so $k = K-1$, $r = k$, and $Y(\lambda) = \Delta_k = \Delta_{k-1} \cup D_k$, i.e. $\lambda = \lambda_{D_k}$ with $|D_k| = k = r$. $\square$

Corollary 2.6. *Let $n = T_{k-1} + r$, $1 \leq r \leq k$.*

- (a) *If $r = k$ (i.e. $n = T_k$), the only cyclic partition of n is $\delta_k = (k, k-1, \dots, 1)$, and $\mathcal{B}(\delta_k) = \delta_k$.*
- (b) *Every cyclic $\mu \vdash n$ satisfies $Y(\mu) \subseteq \Delta_k$, $\Delta_{k-1} \subseteq Y(\mu)$, $\ell(\mu) \in \{k-1, k\}$ and $\mathcal{B}^k(\mu) = \mu$.*
- (c) *If $\mu \vdash n$ is cyclic, then for every $a \geq 0$: $\sum_{i=a}^{a+k-1} \ell(\mathcal{B}^i(\mu)) = k(k-1) + r$. If moreover $r < k$, both values $k-1$ and k occur among $\ell(\mathcal{B}^a(\mu)), \dots, \ell(\mathcal{B}^{a+k-1}(\mu))$.*

Proof. (a) and (b) are immediate from Theorem 2.5 ($E = D_k$ when $r = k$; $\ell(\lambda_E) = k-1 + [d_k(k) \in E]$ was computed in its proof). (c) Write $\mu = \lambda_E$. Then $\mathcal{B}^i(\mu) = \lambda_{\rho^i(E)}$, so $\ell(\mathcal{B}^i(\mu)) = k-1 + [d_k(k) \in \rho^i(E)] = k-1 + [d_k(j_i) \in E]$, where $d_k(j_i) = \rho^{-i}(d_k(k))$, i.e. $j_i \equiv k-i \pmod{k}$ (Lemma 2.1). As i runs through $a, \dots, a+k-1$, j_i runs through all of $1, \dots, k$ once, so the sum is $k(k-1) + |E| = k(k-1) + r$. If $0 < r < k$, some j_i lies in E and some does not. $\square$

3 Piles and the part-count sequence

This is the “ $\text{diagram}_B(\lambda)$ ” bookkeeping of Griggs–Ho [4, §3], written with explicitly labelled piles.

Fix $\lambda \vdash n$, write $\lambda^{(i)} = \mathcal{B}^i(\lambda)$ and

$$c_i = \ell(\lambda^{(i-1)}) \quad (i \geq 1).$$

We call $c(\lambda) = (c_1, c_2, \dots)$ the *part-count sequence*; clearly $c(\lambda^{(a)}) = (c_{a+1}, c_{a+2}, \dots)$. Every $c_i \geq 1$.

Lemma 3.1 (piles). *For every $i \geq 0$ the multiset of parts of $\lambda^{(i)}$ is*

$$\{\lambda_m - i : 1 \leq m \leq \ell(\lambda), \lambda_m > i\} \uplus \{c_j - (i-j) : 1 \leq j \leq i, c_j > i-j\}.$$

Proof. Induction on i ; $i = 0$ is clear. $\lambda^{(i+1)} = \mathcal{B}(\lambda^{(i)})$ has the part $\ell(\lambda^{(i)}) = c_{i+1} = c_{i+1} - 0$ and the parts of $\lambda^{(i)}$ decreased by one, the zeros being discarded: this is the stated multiset for $i+1$ (the term $j = i+1$ is c_{i+1}). $\square$

We use *labels*: $O_1, \dots, O_{\ell(\lambda)}$ (the original piles) and $N_1, N_2, \dots$ (N_j is the pile created by the j -th move, of size c_j when created). By Lemma 3.1, at time i (i.e. in $\lambda^{(i)}$) the pile O_m has size $\lambda_m - i$ and the pile N_j ($j \leq i$) has size $c_j - (i - j)$, a pile being present iff its size is positive. For $i \geq 1$ let R_i be the set of labels of the piles present at time $i - 1$:

$$O_m \in R_i \iff \lambda_m \geq i, \quad N_j \in R_i \iff j \leq i - 1 \text{ and } c_j \geq i - j. \quad (2)$$

By Lemma 3.1 (with $i - 1$), $|R_i| = \ell(\lambda^{(i-1)}) = c_i$. Let $Z_i = R_i \setminus R_{i+1}$ (the piles of size 1 at time $i - 1$, which disappear in move i).

Lemma 3.2 (basic facts). *For all $i \geq 1$:*

- (a) $N_i \in R_{i+1}$ and $N_i \notin R_i$.
- (b) $R_{i+1} = (R_i \setminus Z_i) \cup \{N_i\}$, hence $c_{i+1} = c_i + 1 - |Z_i|$ and $c_{i+1} \leq c_i + 1$.
- (c) If $c_{i+1} = c_i + 1$ then $Z_i = \emptyset$, i.e. $R_i \subseteq R_{i+1}$. If $c_{i+1} = c_i$ then $|Z_i| = 1$.
- (d) For $j < i$: $N_j \in R_i$ implies $N_j \in R_{i'}$ for $j < i' \leq i$, and $N_j \notin R_i$ implies $N_j \notin R_{i'}$ for all $i' \geq i$. Likewise for the labels O_m .

Proof. (a) $c_i \geq 1 = (i + 1) - i$; and labels N_j in R_i have $j \leq i - 1$. (d) For $i' > j$ the clause $j \leq i' - 1$ in (2) holds automatically, so membership of N_j in $R_{i'}$ is the condition $c_j \geq i' - j$, which is monotone in i' ; for O_m the condition $\lambda_m \geq i'$ is monotone as well. (b) By (d), $R_{i+1} \setminus \{N_i\} \subseteq R_i$, so $R_{i+1} = (R_i \cap R_{i+1}) \cup \{N_i\} = (R_i \setminus Z_i) \cup \{N_i\}$, a disjoint union by (a); take cardinalities. (c) follows from $c_{i+1} = c_i + 1 - |Z_i|$. $\square$

Lemma 3.3 (gap lemma). *Let $1 \leq j < i$. If $N_j \notin R_i$ and $N_{j+1} \in R_{i+1}$, then $c_j = i - j - 1$, $c_{j+1} = i - j$, and $N_{j+1} \notin R_{i+2}$.*

Proof. By (2), $c_j \leq i - j - 1$ and $c_{j+1} \geq (i + 1) - (j + 1) = i - j$. By Lemma 3.2(b), $c_{j+1} \leq c_j + 1 \leq i - j$. Hence both equalities hold, and $c_{j+1} = i - j < (i + 2) - (j + 1)$ gives $N_{j+1} \notin R_{i+2}$. $\square$

Lemma 3.4 (the sequence determines the partition). *For $i \geq 1$, $\#\{m : \lambda_m \geq i\} = c_i - \#\{j : 1 \leq j \leq i - 1, c_j \geq i - j\}$. Consequently, if two partitions have the same part-count sequence, they are equal.*

Proof. The formula is $|R_i| = c_i$ together with (2). The numbers $\#\{m : \lambda_m \geq i\}$, $i \geq 1$, determine λ , since $\lambda_m = \#\{i \geq 1 : \#\{m' : \lambda_{m'} \geq i\} \geq m\}$. $\square$

Lemma 3.5 (window sums; generalising [4, Prop. 3.2(2)]). *For all $a \geq 0$ and $h \geq 1$: $c_{a+1} + c_{a+2} + \dots + c_{a+h} \leq n + T_{h-1}$.*

Proof. Let $\mu = \lambda^{(a)}$; its part-count sequence is $(c_{a+1}, c_{a+2}, \dots)$. Applying (2) to μ , for $1 \leq g \leq h$ the number c_{a+g} equals $\#\{m : \mu_m \geq g\}$ plus the number of piles created in the first $g - 1$ moves starting from μ that are still present, which is at most $g - 1$. Summing, $\sum_{g=1}^h c_{a+g} \leq \sum_{g \geq 1} \#\{m : \mu_m \geq g\} + \sum_{g=1}^h (g - 1) = |\mu| + T_{h-1} = n + T_{h-1}$. $\square$

4 Ramps

The following lemmas are Lemmas 3.4–3.6 and Proposition 3.2(3) of Griggs–Ho [4]; we give complete proofs (the proof of [4, Lemma 3.5] ends with “continue this process”, which we replace by an explicit induction) and we add the equality statement in Proposition 4.5.

Definition 4.1. A *ramp* of λ is a triple (p, q, x) of integers with $p \geq 1$, $q \geq p + 2$ and

$$c_p = x - 1, \quad c_{p+1} = c_{p+2} = \cdots = c_{q-1} = x, \quad c_q = x + 1.$$

Its *length* is $q - p$ and its *value* is x (note $x \geq 2$, since $c_p \geq 1$).

Lemma 4.2 (sandwich). *If $1 \leq i < j$ and $c_i < x < c_j$ for an integer x , then there is a ramp (p, q, x) with $i \leq p < q \leq j$.*

Proof. Let p be the largest m with $i \leq m < j$ and $c_m \leq x - 1$ (it exists: $m = i$ qualifies). By Lemma 3.2(b), $c_{p+1} \leq c_p + 1 \leq x$. Since $c_j > x$, this rules out $p + 1 = j$; so $p + 1 < j$, and $c_{p+1} \geq x$ by maximality of p . Hence $c_p = x - 1$ and $c_{p+1} = x$. Let q be the least m with $p < m \leq j$ and $c_m \geq x + 1$ (it exists: $m = j$ qualifies). Then $q \geq p + 2$ since $c_{p+1} = x$. For $p < m < q$ we have $c_m \leq x$ (minimality of q) and $c_m \geq x$ (maximality of p , as $m < j$), so $c_m = x$. Finally $x + 1 \leq c_q \leq c_{q-1} + 1 = x + 1$. $\square$

Lemma 4.3 (short ramps [4, Lemma 3.6]). *If $(p, p + 2, x)$ is a ramp, then $p \leq x$.*

Proof. Suppose $p \geq x + 1$; then $p \geq 3$. Since $|R_p| = c_p = x - 1 \leq p - 2$, not all of $N_1, \dots, N_{p-1}$ lie in R_p , while $N_{p-1} \in R_p$ (Lemma 3.2(a)). Let j be the largest index in $[1, p - 2]$ with $N_j \notin R_p$; then $N_{j+1} \in R_p$. As $c_{p+1} = c_p + 1$ and $c_{p+2} = c_{p+1} + 1$, Lemma 3.2(c) gives $R_p \subseteq R_{p+1} \subseteq R_{p+2}$, so $N_{j+1} \in R_{p+1}$ and $N_{j+1} \in R_{p+2}$. But Lemma 3.3 (with $i = p$) gives $N_{j+1} \notin R_{p+2}$, a contradiction. $\square$

Lemma 4.4 (descent [4, Lemma 3.5]). *Let (p, q, x) be a ramp with $q - p \geq 3$ and $p \geq x + 1$. Then there is a ramp (p', q', x') with*

$$p - x \leq p' \leq p - 2, \quad x' = p - p' \leq x, \quad 2 \leq q' - p' \leq q - p - 1.$$

Proof. The x labels $N_{p-x}, \dots, N_{p-1}$ exist since $p - x \geq 1$. As $|R_p| = x - 1$, one of them is not in R_p ; $N_{p-1} \in R_p$. Let p' be the largest index in $[p - x, p - 2]$ with $N_{p'} \notin R_p$ and put $y = p - p'$, so $2 \leq y \leq x$ and $N_{p'+1}, \dots, N_{p-1} \in R_p$. As $c_{p+1} = c_p + 1$, $R_p \subseteq R_{p+1}$, and with Lemma 3.2(a), $N_{p'+1}, \dots, N_p \in R_{p+1}$. Lemma 3.3 ($j = p'$, $i = p$) gives $c_{p'} = y - 1$ and $c_{p'+1} = y$.

For $1 \leq l \leq q - p - 1$ let $H(l)$ be the statement

$$H(l) : \quad c_{p'+1} = \cdots = c_{p'+l} = y \quad \text{and} \quad N_{p'+l}, N_{p'+l+1}, \dots, N_{p+l-1} \in R_{p+l}.$$

We just proved $H(1)$. Assume $H(l)$. Since $c_{p'+l} = y = p - p'$, (2) gives, for $i > p' + l$: $N_{p'+l} \in R_i \iff i \leq p + l$. Hence $N_{p'+l} \in Z_{p+l}$.

- If $l = q - p - 1$, then $p + l = q - 1$ and $c_q = c_{q-1} + 1$, so $Z_{q-1} = \emptyset$ by Lemma 3.2(c): a contradiction. So $H(q - p - 1)$ is false.
- If $l \leq q - p - 2$, then $p + l$ and $p + l + 1$ lie in $[p + 1, q - 1]$, so $c_{p+l+1} = c_{p+l} = x$ and $|Z_{p+l}| = 1$, i.e. $Z_{p+l} = \{N_{p'+l}\}$. By Lemma 3.2(b), $R_{p+l+1} = (R_{p+l} \setminus \{N_{p'+l}\}) \cup \{N_{p+l}\} \supseteq \{N_{p'+l+1}, \dots, N_{p+l}\}$. In particular $N_{p'+l+1} \in R_{p+l+1}$, i.e. $c_{p'+l+1} \geq (p + l + 1) - (p' + l + 1) = y$, and $c_{p'+l+1} \leq c_{p'+l} + 1 = y + 1$. So either $c_{p'+l+1} = y + 1$, or $c_{p'+l+1} = y$ and then $H(l + 1)$ holds.

Let l_0 be the largest $l \leq q - p - 1$ such that $H(1), \dots, H(l)$ all hold; then $1 \leq l_0 \leq q - p - 2$, and since $H(l_0 + 1)$ fails, the second item gives $c_{p'+l_0+1} = y + 1$. Thus $(p', p' + l_0 + 1, y)$ is a ramp of length $l_0 + 1 \in [2, q - p - 1]$, and $x' = y$ has the stated properties. $\square$

Proposition 4.5 (ramp bound). *Every ramp (p, q, x) satisfies $p \leq (q - p - 1)x$. If equality holds, then $(jx, jx + j + 1, x)$ is a ramp for every $j = 1, 2, \dots, q - p - 1$.*

Proof. Strong induction on the length $L = q - p \geq 2$. If $L = 2$, Lemma 4.3 gives $p \leq x$; if $p = x$, the ramp itself is $(x, x + 2, x)$. Let $L \geq 3$. If $p \leq x$ then $p \leq x < (L - 1)x$. Otherwise Lemma 4.4 provides a ramp (p', q', x') with $p' \geq p - x$, $x' \leq x$ and length $L' \leq L - 1$; by induction $p' \leq (L' - 1)x'$, hence

$$p \leq p' + x \leq (L' - 1)x' + x \leq (L - 2)x + x = (L - 1)x.$$

If $p = (L - 1)x$, all these inequalities are equalities: $p' = p - x = (L - 2)x$ and $(L' - 1)x' = (L - 2)x$; since $1 \leq L' - 1 \leq L - 2$ and $x' \leq x$, this forces $L' = L - 1$ and $x' = x$. So $((L - 2)x, (L - 2)x + L - 1, x)$ is a ramp attaining equality, and by induction $(jx, jx + j + 1, x)$ is a ramp for $j = 1, \dots, L - 2$; for $j = L - 1$ it is (p, q, x) itself. $\square$

Lemma 4.6 (long ramps [4, Lemma 3.4]). *If (p, q, x) is a ramp with $q - p = x + 1$, then $q \leq n + 1$. (Recall that $x \geq 2$ for every ramp.)*

Proof. For $p + 1 \leq j \leq q - 1$ we have $c_j = x \geq q - j$, so $N_j \in R_q$: these are x piles. Also $c_p = x - 1 < q - p$, so $N_p \notin R_q$. Since $|R_q| = c_q = x + 1$, R_q contains exactly one further label P , and $P = O_m$ for some m , or $P = N_j$ with $1 \leq j \leq p - 1$. If $P = O_m$, then $q \leq \lambda_m \leq n$. If $P = N_1$, then $c_1 \geq q - 1$, and $c_1 = \ell(\lambda) \leq n$. Suppose $P = N_j$ with $2 \leq j \leq p - 1$. Then $N_{j-1} \notin R_q$. As $c_q = c_{q-1} + 1$, $R_{q-1} \subseteq R_q$, so $N_{j-1} \notin R_{q-1}$. Next, $q - 2 \geq p + 1$ (as $x \geq 2$), so $c_{q-1} = c_{q-2} = x$ and $|Z_{q-2}| = 1$; since $c_p = x - 1 = (q - 2) - p$, $N_p \in R_{q-2} \setminus R_{q-1}$, so $Z_{q-2} = \{N_p\}$. As $j - 1 \neq p$, $N_{j-1} \in R_{q-2}$ would give $N_{j-1} \in R_{q-1}$ (Lemma 3.2(b)); hence $N_{j-1} \notin R_{q-2}$, i.e. $c_{j-1} \leq q - j - 2$. But $N_j \in R_q$ gives $c_j \geq q - j$, so $c_j \geq c_{j-1} + 2$, contradicting Lemma 3.2(b). $\square$

5 The cyclic partitions of $T_{k-1} + 1$ and the entry lemma

Let $k \geq 3$ and $n = T_{k-1} + 1$ (rank k , $r = 1$). For $1 \leq j \leq k$ put $\zeta_j = \lambda_{\{d_k(j)\}}$: its i -th part is $k - i + [i = j]$ ($1 \leq i \leq k$, zero parts omitted), i.e. ζ_j is δ_{k-1} with 1 added to its j -th part (the k -th part of δ_{k-1} being 0). By Theorem 2.5:

$$\begin{aligned} \text{the cyclic partitions of } n \text{ are exactly } \zeta_1, \dots, \zeta_k; \quad \mathcal{B}(\zeta_j) = \zeta_{j \oplus 1}; \\ \ell(\zeta_j) = k - 1 \quad (1 \leq j \leq k - 1), \quad \ell(\zeta_k) = k, \end{aligned} \tag{3}$$

where $j \oplus 1 = j + 1$ for $j < k$ and $k \oplus 1 = 1$. (Indeed $\rho(d_k(j)) = d_k(j \oplus 1)$ by Lemma 2.1.) The parts of ζ_j for $j \leq k - 1$ are $(k - 1 + [j = 1], k - 2 + [j = 2], \dots, 1 + [j = k - 1])$, and $\zeta_k = (k - 1, k - 2, \dots, 1, 1)$.

Lemma 5.1 (preimages). *Let ζ be a partition and $s \geq 1$. There is at most one partition μ with $\ell(\mu) = s$ and $\mathcal{B}(\mu) = \zeta$. It exists iff s is a part of ζ and $s \geq \ell(\zeta) - 1$. Its parts are then the numbers $z + 1$ for z in the multiset $\zeta \ominus \{s\}$ (ζ with one copy of s removed), together with $s - \ell(\zeta) + 1$ parts equal to 1.*

Proof. If $\mathcal{B}(\mu) = \zeta$ and $\ell(\mu) = s$, then by definition the parts of ζ are s and the $\mu_m - 1$ with $\mu_m \geq 2$. So s is a part of ζ , and $\{\mu_m - 1 : \mu_m \geq 2\} = \zeta \ominus \{s\}$ has $\ell(\zeta) - 1$ elements. The remaining $s - \ell(\zeta) + 1 \geq 0$ parts of μ equal 1. This determines μ . Conversely, the μ so described has s parts and $\mathcal{B}(\mu) = \zeta$. $\square$

Lemma 5.2 (entry lemma). *Let $k \geq 3$, $n = T_{k-1} + 1$, $\lambda \vdash n$ and $d = d_{\mathcal{B}}(\lambda) \geq 1$. Then $\lambda^{(d)} = \zeta_j$ for some $j \in \{1, 3, 4, \dots, k-1\}$, and $c_d = \ell(\lambda^{(d-1)}) = k-2$.*

Proof. $\mu = \lambda^{(d-1)}$ is not cyclic, and $\mathcal{B}(\mu) = \lambda^{(d)} = \zeta_j$ for some j by (3). Let $s = \ell(\mu)$. By (3), the cyclic partition $\zeta_{j \ominus 1}$ ($j \ominus 1 = j-1$ for $j \geq 2$, $1 \ominus 1 = k$) also satisfies $\mathcal{B}(\zeta_{j \ominus 1}) = \zeta_j$. So by the uniqueness in Lemma 5.1, $s \neq \ell(\zeta_{j \ominus 1})$.

- $j = k$: $s \geq \ell(\zeta_k) - 1 = k-1$ and s is a part of ζ_k , all of whose parts are $\leq k-1$; so $s = k-1 = \ell(\zeta_{k-1})$, which is excluded.
- $j = 1$: $s \geq k-2$. The parts of $\zeta_1 = (k, k-2, k-3, \dots, 1)$ that are $\geq k-2$ are k and $k-2$; $s = k = \ell(\zeta_k)$ is excluded, so $s = k-2$.
- $2 \leq j \leq k-1$: $s \geq k-2$, all parts of ζ_j are $\leq k-1$, and $s = k-1 = \ell(\zeta_{j-1})$ is excluded; so $s = k-2$. For $j = 2$, $\zeta_2 = (k-1, k-1, k-3, \dots, 1)$ has no part $k-2$, which is impossible. So $j \neq 2$.

Hence $j \in \{1, 3, \dots, k-1\}$ and $c_d = s = k-2$. □

6 The upper bound

Theorem 6.1. *Let $k \geq 5$ and $\lambda \vdash T_{k-1} + 1$. Then $d_{\mathcal{B}}(\lambda) \leq (k-1)(k-3)$.*

Proof. Let $d = d_{\mathcal{B}}(\lambda)$; assume $d \geq 1$. By Lemma 5.2, $c_d = k-2$ and $\lambda^{(d)} = \zeta_j$ with $j \in \{1, 3, \dots, k-1\}$. For $m \geq 0$, $c_{d+1+m} = \ell(\lambda^{(d+m)}) = \ell(\mathcal{B}^m(\zeta_j))$, which by (3) equals k if $j+m \equiv 0 \pmod{k}$ and $k-1$ otherwise. So $c_{d+1+m} = k-1$ for $0 \leq m \leq k-j-1$ and $c_{d+1+(k-j)} = k$. That is, $(d, d+1+k-j, k-1)$ is a ramp of length $k-j+1 \geq 2$.

- If $3 \leq j \leq k-1$: Proposition 4.5 gives $d \leq (k-j)(k-1) \leq (k-3)(k-1)$.
- If $j = 1$: the ramp has length $k = (k-1) + 1$, so Lemma 4.6 (with $x = k-1 \geq 2$) gives $d+k \leq n+1$, i.e. $d \leq T_{k-1} + 2 - k$. And $(k-1)(k-3) - (T_{k-1} + 2 - k) = \frac{1}{2}(k^2 - 5k + 2) > 0$ for $k \geq 5$.

□

7 The lower bound: the Griggs–Ho witness

For $k \geq 4$, $h \in \{1, \dots, k-1\}$ and $a \in \{1, \dots, k\}$ let (with $\oplus_w$ as before: $j \oplus_w 1 = j+1$ for $j < w$, $w \oplus_w 1 = 1$)

$$Z(h, a) = \Delta_{k-2} \cup (D_{k-1} \setminus \{d_{k-1}(h)\}) \cup \{d_k(a), d_k(a \oplus_k 1)\}.$$

Lemma 7.1. *Let $k \geq 4$ and let μ be a partition with $Y(\mu) = Z(h, a)$. If $(h, a) \neq (k-1, 1)$, then $Y(\mathcal{B}(\mu)) = Z(h \oplus_{k-1} 1, a \oplus_k 1)$. If $(h, a) = (k-1, 1)$, then $\mu = (k, k-1, k-3, k-4, \dots, 2)$ and $\mathcal{B}(\mu) = \zeta_3$.*

Proof. Let $A = \{a, a \oplus_k 1\}$. Counting the cells $d_w(w)$ of row 1 and $d_w(1)$ of column 1: $\ell(\mu) = k-2 + [h \neq k-1] + [k \in A]$ and $\mu_1 = k-2 + [h \neq 1] + [1 \in A]$. So $\ell(\mu) \leq \mu_1 - 2$ iff $h = k-1$, $k \notin A$, $h \neq 1$ and $1 \in A$. Now $1 \in A$ iff $a \in \{1, k\}$, and $k \notin A$ excludes $a \in \{k-1, k\}$. So this happens iff $(h, a) = (k-1, 1)$ (and then $h \neq 1$ as $k \geq 4$). Otherwise Corollary 2.4 and Lemma 2.1 give $Y(\mathcal{B}(\mu)) = \rho(Z(h, a)) = Z(h \oplus_{k-1} 1, a \oplus_k 1)$. If $(h, a) = (k-1, 1)$, the cells outside Δ_{k-2}

are $d_{k-1}(1), \dots, d_{k-1}(k-2)$, $d_k(1)$ and $d_k(2)$. So column 1 has $(k-2) + 1 + 1 = k$ cells, column 2 has $(k-3) + 1 + 1 = k-1$, column $j \in [3, k-2]$ has $(k-1-j) + 1 = k-j$, and columns $k-1, k$ are empty. Thus $\mu = (k, k-1, k-3, \dots, 2)$ with $k-2$ parts. $\mathcal{B}(\mu)$ has the parts $k-2$ and $k-1, k-2, k-4, k-5, \dots, 1$, i.e. $\mathcal{B}(\mu) = (k-1, k-2, k-2, k-4, \dots, 1) = \zeta_3$. $\square$

Theorem 7.2. *For $k \geq 4$, $\lambda^{\{k\}} = (k-2, k-2, k-3, \dots, 3, 2, 2, 1)$ is a partition of $T_{k-1} + 1$ with $d_{\mathcal{B}}(\lambda^{\{k\}}) = (k-1)(k-3)$.*

Proof. The sum is $(k-2) + (T_{k-2} - 1) + 2 + 1 = T_{k-1} + 1$. Its diagram is $Z(1, k-1)$. Column 1 has $k-2$ cells (those of Δ_{k-2} ; $d_{k-1}(1)$ missing). Column 2 has $(k-3) + 1 = k-2$ (Δ_{k-2} and $d_{k-1}(2)$). Column $j \in [3, k-2]$ has $(k-1-j) + 1 = k-j$. Column $k-1$ has $d_{k-1}(k-1) = (1, k-1)$ and $d_k(k-1) = (2, k-1)$, so 2 cells. Column k has $d_k(k) = (1, k)$, so 1 cell.

For $m \geq 0$ let $h_m \in [1, k-1]$ with $h_m \equiv 1+m \pmod{k-1}$ and $a_m \in [1, k]$ with $a_m \equiv k-1+m \pmod{k}$. Then $(h_m, a_m) = (k-1, 1)$ iff $m \equiv -1 \pmod{k-1}$ and $m \equiv 2 \pmod{k}$. $M = k^2 - 4k + 2$ satisfies both (modulo $k-1$, $k \equiv 1$ gives $M \equiv -1$), and for $k \geq 4$, $0 \leq M < k(k-1)$. So M is the least non-negative solution. By Lemma 7.1 and induction, $Y(\mathcal{B}^m(\lambda^{\{k\}})) = Z(h_m, a_m)$ for $0 \leq m \leq M$, and $\mathcal{B}^{M+1}(\lambda^{\{k\}}) = \zeta_3$ is cyclic. For $m \leq M$ the diagram misses a cell of D_{k-1} , while every cyclic partition of n contains Δ_{k-1} (Corollary 2.6(b)); so $\mathcal{B}^m(\lambda^{\{k\}})$ is not cyclic. Hence $d_{\mathcal{B}}(\lambda^{\{k\}}) = M + 1 = k^2 - 4k + 3 = (k-1)(k-3)$. $\square$

Corollary 7.3. *$D_{\mathcal{B}}(T_{k-1} + 1) = (k-1)(k-3)$ for every $k \geq 5$. (For $k = 4$, $D_{\mathcal{B}}(7) = 4 > 3$; for $k = 3$, $D_{\mathcal{B}}(4) = 2$; these side remarks are exhaustive computations, Appendix A, and are not used anywhere.)*

8 All extremal partitions

Theorem 8.1. *Let $k \geq 5$ and $\lambda \vdash T_{k-1} + 1$. The following are equivalent:*

- (1) $d_{\mathcal{B}}(\lambda) = (k-1)(k-3)$;
- (2) $c_{(k-1)(k-3)} \leq k-2$, i.e. $\mathcal{B}^{(k-1)(k-3)-1}(\lambda)$ has at most $k-2$ parts;
- (3) $(j(k-1), j(k-1) + j + 1, k-1)$ is a ramp for every $j = 1, \dots, k-3$.

In this case $\mathcal{B}^{(k-1)(k-3)}(\lambda) = \zeta_3$.

Proof. (3) $\Rightarrow$ (2): $j = k-3$ gives $c_{(k-3)(k-1)} = k-2$. (2) $\Rightarrow$ (1): a partition with at most $k-2$ parts is not cyclic, by (3). So $d_{\mathcal{B}}(\lambda) \geq (k-1)(k-3)$, and equality holds by Theorem 6.1. (1) $\Rightarrow$ (3): in the proof of Theorem 6.1, $j = 1$ gives $d < (k-1)(k-3)$, and $j \geq 4$ gives $d \leq (k-4)(k-1)$. So $j = 3$, and the ramp $(d, d+k-2, k-1)$ has length $L = k-2$ and satisfies $d = (L-1)(k-1)$. The equality part of Proposition 4.5 gives (3); and $\lambda^{(d)} = \zeta_j = \zeta_3$. $\square$

Numbers of extremal partitions (`bs.py`):

k	5	6	7	8	9	10	11
$\#\{\lambda \vdash T_{k-1} + 1 : d_{\mathcal{B}}(\lambda) = (k-1)(k-3)\}$	3	12	62	288	1310	5862	26399

9 Where the Cell 3 argument loses, and the extra idea

What the Cell 3 argument gives. That argument (Griggs–Ho [4, Thm. 4.4]) is valid for every $r \in [1, k-1]$ and yields $k^2 - 2k - 1$. For $r = 1$ this exceeds the truth $(k-1)(k-3)$ by $2k-4$. It loses in two places.

1. *The anchor.* It bounds $d \leq t-1$, where t is the first time at which the periodic regime shows the pattern $(k-1, k)$ of part counts. For $r = 1$ the cycle has a single k per period: if the orbit enters the cycle at ζ_j , the first k appears only $k-j$ steps later, so $t-1 = d+k-j-1$. Up to $k-2$ steps are thrown away.
2. *The value of the ramp.* It cannot exclude ramps of value k (case (A) of the tail lemma). For these the ramp bound gives up to $(k-3)k+k-1 = k^2-2k-1$, and this is exactly where its final bound comes from.

A cheap improvement comes from comparison (Remark 9.1): $D_{\mathcal{B}}(T_{k-1}+1) \leq D_{\mathcal{B}}(T_{k-1}) = (k-1)(k-2)$. This is still $k-1$ too large.

The extra idea: analyse the entry into the cycle exactly. Lemma 5.2 classifies the last non-cyclic partition $\lambda^{(d-1)}$ by counting preimages (Lemma 5.1). It always has exactly $k-2$ parts, and the entry point is ζ_j with $j \neq 2, k$. This removes both losses at once. There is a ramp starting exactly at $p = d$ (no anchor loss), and its value is $k-1$, not k . The ramp bound then reads $d \leq (k-j)(k-1)$, where the ramp has length $k-j+1$. The only entries that could beat $(k-1)(k-3)$ are $j = 2$ (length $k-1$, giving $(k-2)(k-1)$, the comparison bound) and $j = 1$ (length k). The first cannot occur: ζ_2 has no part $k-2$, and its only preimage is the cyclic ζ_1 . The second is handled by the long-ramp lemma, Lemma 4.6. So the $k-1$ steps between the comparison bound and the truth are exactly the exclusion of entry at ζ_2 .

Remark 9.1 (comparison, not used above). If $\lambda_i \leq \mu_i$ for all i , then $\mathcal{B}(\lambda)_i \leq \mathcal{B}(\mu)_i$ for all i (Akin–Davis [1, Thm. 3(c)], [4, Thm. 4.1]). Proof: let $x = (\ell(\lambda), \max(\lambda_1-1, 0), \max(\lambda_2-1, 0), \dots)$ and define y likewise for μ . Then $x_i \leq y_i$ for all i , since $\ell(\lambda) \leq \ell(\mu)$. For $v \geq 1$, $\mathcal{B}(\lambda)_i \geq v$ iff $\#\{j : x_j \geq v\} \geq i$, and $\#\{j : x_j \geq v\} \leq \#\{j : y_j \geq v\}$. Hence $\mathcal{B}(\lambda)_i \leq \mathcal{B}(\mu)_i$.

Now let $\lambda \vdash T_{k-1}+1$, and let $\nu \vdash T_{k-1}$ be obtained by lowering the last part of λ by one. By induction $\mathcal{B}^m(\nu)_i \leq \mathcal{B}^m(\lambda)_i$ for all i . For $m = (k-1)(k-2)$, Cell 2 gives $d_{\mathcal{B}}(\nu) \leq m$, and δ_{k-1} is the only cyclic partition of T_{k-1} (Corollary 2.6(a)), so $\mathcal{B}^m(\nu) = \delta_{k-1}$. So $\mathcal{B}^m(\lambda)$ is obtained from δ_{k-1} by adding 1 to a single part i (possibly a zero part). Being non-increasing, it has $i \leq k$, i.e. $\mathcal{B}^m(\lambda) = \zeta_i$, which is cyclic. Hence $d_{\mathcal{B}}(\lambda) \leq (k-1)(k-2)$. This bound is weaker than Theorem 6.1 and is not used there.

Outlook (observations only, nothing claimed). For $1 \leq r < \lfloor (k-1)/2 \rfloor$ the Griggs–Ho lower bound $(k-3-r)k+r+2$ equals $(k-1)(k-2-r)$. This is exactly the ramp bound for a value- $(k-1)$ ramp of length $k-1-r$ starting at d , which suggests the same mechanism. For $r \geq 2$, however, the analogue of Lemma 5.2 fails: computations show orbits whose last non-cyclic partition has $k-1$ parts, and then no ramp starts at d . For instance, for $n = T_{k-1}+2$ with $k = 7, 8, 9$, such orbits reach $d = 17, 23, 30$ (computed by `bs.py`: Cell 5 section of its output `bs_output_60.txt`, maximum of $d_{\mathcal{B}}$ over entry type $E = \{1, 2\}$, the only type whose last non-cyclic partition has $k-1$ parts). Those values are below the conjectured $(k-1)(k-4) = 18, 28, 40$, but they need a different argument. So the method alone proves the case $r = 1$ but does not extend to $r \geq 2$. Cell 5 adds the missing ingredient for $r = 2$ (a pile-chain lemma bounding those entries by about T_{k-2}) and proves $D_{\mathcal{B}}(T_{k-1}+2) = (k-1)(k-4)$ for $k \geq 7$.

10 Computations

No computation is used in the proofs above. As sanity checks, `bs.py` (exhaustive for $n \leq 60$) confirms: $D_{\mathcal{B}}(T_{k-1} + 1) = (k-1)(k-3)$ for $5 \leq k \leq 11$; $d_{\mathcal{B}}(\lambda^{\{k\}}) = (k-1)(k-3)$ for $5 \leq k < 80$; Theorem 8.1 for $5 \leq k \leq 10$. A separate exhaustive check for $5 \leq k \leq 9$ (`bs.py`, section “C4 entry lemma” of its output `bs_output_60.txt`) confirms the statement of Lemma 5.2, and that the maximum of $d_{\mathcal{B}}$ over orbits entering at ζ_j equals $(k-j)(k-1)$ for $j \geq 3$ and $T_{k-1} + 2 - k$ for $j = 1$, so every case of the proof is sharp. It also confirms $D_{\mathcal{B}}(T_{k-1} + 2) = (k-1)(k-4)$ for $7 \leq k \leq 11$ (proved for all $k \geq 7$ in Cell 5).

A Code for the exhaustive small cases

File `p3_small_cases.py` (Python 3, no dependencies; runs in well under one second). It computes $d_{\mathcal{B}}$ directly from the definition: the orbit of λ is followed until a partition repeats, and the index of the first occurrence of that partition is $d_{\mathcal{B}}(\lambda)$.

```
#!/usr/bin/env python3
"""Exhaustive small cases used in p3_c2/p3_c3/p3_c4/p3_c5.
Straight from the definitions: B(lambda), d_B(lambda) = number of steps before the
first partition that recurs in the orbit, D_B(n) = max over all partitions of n.
Runtime: well under one second."""

def partitions(n, maxp=None):
    maxp = n if maxp is None else maxp
    if n == 0:
        yield ()
        return
    for p in range(min(n, maxp), 0, -1):
        for rest in partitions(n - p, p):
            yield (p,) + rest

def B(lam):
    return tuple(sorted([len(lam)] + [x - 1 for x in lam if x > 1], reverse=True))

def d_B(lam):
    first = {}
    i = 0
    while lam not in first:
        first[lam] = i
        lam = B(lam)
        i += 1
    return first[lam] # the tail length: B^i(lam) is cyclic iff i >= this

def fmt(lam):
    # compact notation: (3,3,2,1) -> 3^2 2 1
    out = []
    for v in sorted(set(lam), reverse=True):
        m = lam.count(v)
        out.append(f"{v}^{m}" if m > 1 else f"{v}")
    return "(" + " ".join(out) + ")"

for n in [1, 3, 4, 5, 6, 7, 8, 9, 12, 17]:
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parts = list(partitions(n))
d = {lam: d_B(lam) for lam in parts}
D = max(d.values())
ext = [lam for lam in parts if d[lam] == D]
print(f"n={n}: {len(parts)} partitions, D_B={D}, extremal: {'', '.join(fmt(l) for l in ext)}")

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Output:

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n=1: 1 partitions, D_B=0, extremal: (1)
n=3: 3 partitions, D_B=2, extremal: (1^3)
n=4: 5 partitions, D_B=2, extremal: (1^4)
n=5: 7 partitions, D_B=3, extremal: (1^5)
n=6: 11 partitions, D_B=6, extremal: (2^2 1^2)
n=7: 15 partitions, D_B=4, extremal: (1^7)
n=8: 22 partitions, D_B=5, extremal: (1^8)
n=9: 30 partitions, D_B=7, extremal: (3 2^2 1^2)
n=12: 77 partitions, D_B=8, extremal: (3^2 2^2 1^2), (1^12)
n=17: 297 partitions, D_B=12, extremal: (1^17)

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References

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